

Knowledge Distillation Across Teacher Architectures for Edge-Efficient Plant Disease Classification

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Abstract. Accurate image-based plant disease classification can support precision agriculture, but high-capacity networks are not always convenient for edge or mobile deployment. This work studies knowledge distillation from large teacher networks into a compact MobileNetV3-Small student, and asks in particular whether the *architecture of the teacher* affects the distilled student. As a preliminary demonstration, we distill a ResNet50 teacher into MobileNetV3-Small on the 38-class PlantVillage dataset with a single train/validation/test split; the distilled student reaches 0.9937 accuracy and 0.9907 macro-F1 with 1.56M parameters, about 15 times fewer than the teacher. We then carry out the main study on the smaller and class-imbalanced Cassava leaf-disease dataset (five classes) using stratified five-fold cross-validation. Three teachers from different architectural families --- ResNet50 (residual CNN), EfficientNet-B2 (compound-scaled CNN), and ViT-B/16 (vision transformer) --- are each retrained per fold and distilled into the same student, and we report accuracy together with macro-averaged precision, recall, and F1. On Cassava, distillation improves the student over a supervised baseline by 1.4--1.8 percentage points of accuracy, but the choice of teacher has little effect: the three distilled students agree to within 0.5 points despite the teachers differing widely in accuracy (82.4--85.1%) and in size (7.7M--85.8M parameters). Strikingly, the weakest and largest teacher (ViT-B/16) yields a student on par with those of the stronger convolutional teachers, and one that even exceeds its own teacher's accuracy. These results suggest that, in this setting, the benefit of distillation comes from the procedure itself rather than from the capacity or architecture of the teacher.

Keywords: Knowledge distillation · Plant disease classification · MobileNetV3 · Cross-validation · Vision transformer · Edge inference.

1 Introduction

Plant disease recognition is a practical computer vision problem with direct relevance to crop monitoring and precision agriculture. Deep neural networks can achieve strong classification accuracy on curated leaf image datasets, but

models with high parameter counts may be inconvenient for mobile or edge-oriented deployment. This creates a tension between predictive accuracy and computational efficiency.

This work studies that trade-off through knowledge distillation, in which a high-capacity teacher network guides the training of a lightweight MobileNetV3-Small student. The aim is not to claim deployment on a specific device, but to evaluate whether a compact student can retain a teacher’s predictive behavior while using far fewer parameters. Beyond the standard teacher-to-student comparison, we focus on a question that a single experiment cannot answer: does the *choice of teacher architecture* change how well the student learns? To study this we compare three teachers from distinct design families under a cross-validated protocol.

The contributions are as follows. First, as a preliminary demonstration, we distill a ResNet50 teacher into MobileNetV3-Small on the 38-class PlantVillage dataset using a single split. Second, and as the main study, we evaluate three teacher architectures --- ResNet50, EfficientNet-B2, and ViT-B/16 --- distilled into the same student on the Cassava leaf-disease dataset. Third, we use a methodologically careful protocol: stratified five-fold cross-validation with the teacher retrained from scratch on each fold’s training partition, so that the held-out fold never influences the teacher. Fourth, we report macro-averaged precision and recall alongside accuracy and macro-F1, which is important given the class imbalance of the Cassava data, present results as mean $\pm$ standard deviation across folds, and assess each distilled student against the supervised baseline with a paired significance test. Finally, we apply post-training quantization to the distilled student as an additional edge-compression step and analyze the resulting accuracy/size trade-off.

2 Background and Related Work

PlantVillage provides a widely used collection of labeled plant leaf images for plant health research [8]. The dataset has enabled controlled studies of image-based plant disease classification, although its curated imaging conditions are cleaner than many field scenarios. The Cassava leaf-disease data used in the main study [12] consists of five classes (four disease categories and a healthy class) and is more class-imbalanced, providing a harder and less controlled testbed. These two datasets were chosen deliberately: PlantVillage is a clean, widely used benchmark that is well suited to an initial controlled demonstration of the distillation pipeline, whereas Cassava consists of real competition field imagery with strong class imbalance, so it tests whether the same conclusions hold under more realistic and challenging conditions. Both are also directly relevant to the edge-deployment motivation, since plant disease diagnosis is a natural use case for on-device inference in the field.

Deep learning has been applied extensively to plant disease classification. Early work showed that convolutional networks trained on PlantVillage can classify leaf diseases with high accuracy [10], and later studies compared fine-tuning

strategies across architectures for this task [14]. Most of this literature focuses on maximizing accuracy with large backbones; comparatively little attention has been paid to compressing these models for on-device inference, which is the gap this work addresses through distillation.

Knowledge distillation was introduced as a way to transfer information from a larger teacher model to a smaller student model [6]. Instead of learning only from one-hot labels, the student also learns from the teacher’s softened class probabilities, which can encode similarities between classes, making distillation attractive when the deployment model must be smaller than the model used to generate the strongest predictions. Beyond matching output logits, later methods transfer intermediate representations, such as hidden-layer hints [11] and spatial attention maps [16]; a broad overview of these variants is given by Gou et al. [4]. In this work we use the original logit-based formulation, since our focus is the choice of teacher rather than the distillation mechanism.

A recurring and directly relevant finding is that a more accurate teacher does not necessarily produce a better student. Cho and Hariharan [2] show that very large teachers can distill less effectively into small students because of the capacity gap between them, and Mirzadeh et al. [9] introduce intermediate “teacher-assistant” networks to bridge that gap. Beyer et al. [1] further argue that the consistency of the teacher-student training setup matters more than raw teacher accuracy. These results motivate our central question: when the student is fixed, does the teacher’s *architecture* --- and not merely its accuracy --- affect the distilled student? Distillation has also been studied in the context of vision transformers, most notably DeiT [15], which distills *into* a transformer using a CNN teacher. Our setting is complementary: we distill *from* a transformer (and from two CNNs) into a compact convolutional student, comparing teacher families under a single protocol.

The teachers in this work span three architectural families. ResNet50 is a residual network that uses skip connections to ease optimization of deeper convolutional models [5]. EfficientNet-B2 is a convolutional network obtained by compound scaling of depth, width, and resolution, and is known for strong accuracy at a low parameter count [13]. ViT-B/16 is a vision transformer that treats an image as a sequence of patches and relies on self-attention rather than convolution [3], giving it a substantially different inductive bias from the two CNN teachers. The student in all experiments is MobileNetV3-Small, an architecture designed for mobile inference through efficient convolutional blocks and hardware-aware design choices [7]. These models were chosen for a specific reason. The three teachers are well-established, strong image classifiers with pretrained ImageNet weights, but they represent deliberately different design philosophies --- residual convolutions, compound-scaled convolutions, and patch-based self-attention --- and they also differ widely in parameter count, which lets us separate the effect of teacher architecture from that of teacher size. MobileNetV3-Small was chosen as the single student because it is a representative compact model intended for edge and mobile deployment, making it the natural target for compression. Comparing CNN and transformer teachers that distill

into this same convolutional student lets us probe whether teacher architecture, and not only teacher accuracy, influences the student.

3 Methodology

Let z_t and z_s denote the teacher and student logits for an input image, and let y denote the ground-truth class label. Supervised teacher and supervised student baselines are trained with cross-entropy against the hard labels. For distillation, the teacher is first trained and then frozen, and the student is optimized using a weighted sum of the cross-entropy loss and a temperature-scaled Kullback-Leibler divergence term:

$$\mathcal{L} = \alpha T^2 \text{KL} \left(\text{softmax} \left(\frac{z_t}{T} \right) \parallel \text{softmax} \left(\frac{z_s}{T} \right) \right) + (1 - \alpha) \text{CE}(y, z_s). \quad (1)$$

The T^2 factor follows the common practice of compensating for the gradient scaling introduced by softened probabilities. The temperature is fixed at $T = 4.0$ throughout. The distillation weight is $\alpha = 0.5$ in the preliminary PlantVillage experiment and $\alpha = 0.7$ in the main Cassava study; in both cases α is held constant across teachers so that any difference between teachers is attributable to the teacher rather than to the loss weighting. MobileNetV3-Small is used as the student in every experiment because it has substantially fewer parameters than any of the teachers while retaining a convolutional design appropriate for image classification.

4 Experimental Setup

All experiments use pretrained ImageNet weights, images resized to 224×224 pixels and normalized with ImageNet statistics, random horizontal flipping and small random rotations during training, deterministic resizing for validation and test, and the best validation checkpoint for final evaluation. Inference latency is reported from a batch-one timing routine that averages repeated forward passes; latency values are local hardware measurements rather than hardware-independent deployment benchmarks.

4.1 Main Study: Cross-Validated Multi-Teacher Distillation on Cassava

The main study uses the Cassava leaf-disease dataset (five classes) and stratified five-fold cross-validation. The dataset is partitioned into five stratified folds, preserving class proportions. For each fold, that fold serves as the test set, and the remaining data is split (again with stratification) into training and validation. Crucially, the teacher is *retrained from scratch on that fold’s training partition* before being frozen and used to distill the student. This avoids the optimistic bias

that would arise if a single teacher had been exposed to images that later appear in a fold’s test set. The fold partition is fixed by seed, so every teacher and every model sees identical folds. All models are initialized from ImageNet-pretrained weights, so the comparison concerns transfer to the plant-disease task rather than training from random initialization; this pretraining is shared across all teachers and the student and therefore does not bias the teacher comparison.

Three teachers are evaluated --- ResNet50, EfficientNet-B2, and ViT-B/16 --- each distilled into a MobileNetV3-Small student, and compared against a supervised MobileNetV3-Small baseline. The student schedule and distillation hyperparameters ($T = 4.0$, $\alpha = 0.7$) are held constant across teachers. Per-fold teacher schedules and the student schedule are summarized in Table 1. Each teacher is also evaluated on its own, trained supervised under the same cross-validation, so that the teachers’ own cross-validated performance can be reported alongside the students. These runs use an NVIDIA RTX 4070 (CUDA) with automatic mixed precision. For each model we report the mean and standard deviation across the five folds of accuracy and of macro-averaged precision, recall, and F1; macro averaging weights each class equally, which matters because the Cassava classes are imbalanced. Because the folds are shared across models, each distilled student can be compared against the supervised baseline with a paired two-sided t -test on the five per-fold accuracies; we report the resulting p -values so that differences are not over-interpreted given the small number of folds.

Table 1. Per-fold training configuration for the Cassava cross-validation study. All teachers are distilled into the same MobileNetV3-Small student with $T = 4.0$ and $\alpha = 0.7$.

Model	Role	Learning rate	Epochs	Batch
ResNet50	Teacher	3×10^{-4}	10	64
EfficientNet-B2	Teacher	3×10^{-4}	10	64
ViT-B/16	Teacher	1×10^{-4}	10	32
MobileNetV3-Small	Student / baseline	5×10^{-4}	15	64

4.2 Preliminary Study: Single-Split Distillation on PlantVillage

As a preliminary demonstration, a ResNet50 teacher, a supervised MobileNetV3-Small baseline, and a distilled MobileNetV3-Small student are trained on the full 38-class PlantVillage dataset with a single deterministic 70/15/15 train/validation/test split (seed 42). These runs use 15 epochs, weight decay 10^{-4} , batch size 32, and distillation settings $T = 4.0$ and $\alpha = 0.5$, and were executed on Apple Silicon (MPS). The ResNet50 teacher uses learning rate 3×10^{-4} , the supervised student uses 10^{-3} , and the distilled student uses 5×10^{-4} .

5 Results

5.1 Main Study: Cassava Cross-Validation

Table 2 summarizes the cross-validated results on Cassava. The table reports each teacher trained supervised, the supervised MobileNetV3-Small baseline (teacher “--”), and the student distilled from each teacher, as mean $\pm$ standard deviation across the five folds. Two questions are of interest: whether distillation improves the student over the supervised baseline on this harder, imbalanced dataset, and whether a stronger or architecturally different teacher distills better than ResNet50.

Table 2. Cross-validated test results on cassava (5-fold, mean $\pm$ standard deviation across folds). Precision, recall, and F1 are macro-averaged. The teacher is retrained per fold; the student and KD hyperparameters are fixed.

Teacher	Student	Accuracy (%)	Precision (%)	Recall (%)	Macro-F1 (%)	Parameters
--	resnet50	85.13 $\pm$ 0.95	75.08 $\pm$ 1.40	73.69 $\pm$ 1.28	74.00 $\pm$ 0.78	23,518,277
--	efficientnet_b2	84.59 $\pm$ 0.67	73.91 $\pm$ 1.51	71.75 $\pm$ 0.86	72.60 $\pm$ 0.61	7,708,039
resnet50	mobilenet_v3_small	83.49 $\pm$ 0.31	72.70 $\pm$ 0.66	69.74 $\pm$ 2.62	70.72 $\pm$ 1.99	1,522,981
vit_b_16	mobilenet_v3_small	83.12 $\pm$ 0.72	72.90 $\pm$ 0.99	68.25 $\pm$ 2.36	69.85 $\pm$ 1.95	1,522,981
efficientnet_b2	mobilenet_v3_small	83.04 $\pm$ 0.41	71.75 $\pm$ 1.53	69.40 $\pm$ 1.98	70.34 $\pm$ 1.26	1,522,981
--	vit_b_16	82.40 $\pm$ 0.64	71.41 $\pm$ 1.28	69.08 $\pm$ 1.58	69.38 $\pm$ 1.26	85,802,501
--	mobilenet_v3_small	81.67 $\pm$ 1.01	69.83 $\pm$ 2.64	67.57 $\pm$ 1.38	68.22 $\pm$ 0.80	1,522,981

Distillation helps. All three teachers raise the student above the supervised baseline (81.67% accuracy) by a similar margin: +1.82 points for ResNet50 (83.49%), +1.45 for ViT-B/16 (83.12%), and +1.37 for EfficientNet-B2 (83.04%). In the paired t -test the ResNet50 and ViT-B/16 gains are statistically significant ($p = 0.029$ and $p = 0.036$), whereas the EfficientNet-B2 gain does not reach significance ($p = 0.079$); given the consistent positive mean difference, this most likely reflects the limited power of five folds together with that run’s larger fold-to-fold variance rather than an absence of effect. The macro-F1 scores improve in step with accuracy, so the gains are not confined to the majority class.

The more notable result is how little the teacher matters. The three distilled students lie within 0.5 points of one another (83.04–83.49%), a spread no larger than their across-fold standard deviations, even though the teachers differ substantially in cross-validated accuracy (82.40–85.13%), in architectural family (two CNNs and one transformer), and in size (7.7M, 23.5M, and 85.8M parameters). Teacher accuracy is therefore a poor predictor of student accuracy here: the weakest and by far the largest teacher, ViT-B/16, produces a student as good as the one distilled from the strongest teacher, and does so with a statistically significant improvement over the baseline. The ViT-B/16 student (83.12%) in fact surpasses its own teacher (82.40%), which is inconsistent with the view that the student simply inherits the teacher’s accuracy. It also runs counter to the expectation that a very large teacher, separated from a 1.5M-parameter student by a wide capacity gap, would distill poorly. Taken together,

these observations indicate that in this setting the improvement comes from the distillation procedure --- the softened-label regularization --- more than from the specific teacher.

This has a practical implication for the edge-deployment motivation. Because the deployed model is always the same 1.5M-parameter student and the teacher is used only during training, a smaller and cheaper teacher is sufficient: EfficientNet-B2, at roughly a third of ResNet50’s parameters, teaches about as well, and the much larger ViT-B/16 offers no advantage as a teacher despite its cost. The deployment cost is unchanged by the teacher in any case: all four students share the MobileNetV3-Small architecture and therefore have essentially the same batch-one inference latency (approximately 5 ms on the RTX 4070), and the teacher is discarded after training. Reporting mean $\pm$ standard deviation across folds, rather than a single split, makes these comparisons more robust to the particular partition of the data.

5.2 Sensitivity to Distillation Hyperparameters

The main study fixes the distillation temperature and weight at the widely used values $T = 4.0$ and $\alpha = 0.7$, and holds them constant across teachers so that differences are attributable to the teacher. To check that this choice is reasonable and that the comparison is not an artifact of a particular setting, we run a single-split sensitivity analysis for the ResNet50 teacher on fold 0 of the Cassava partition. Because T and α affect only the student loss, the teacher is trained once and reused for every setting; we then sweep $T \in \{1, 2, 4, 8\}$ at $\alpha = 0.7$ and $\alpha \in \{0.3, 0.5, 0.7, 0.9\}$ at $T = 4.0$. Table 3 reports the resulting student accuracy and macro-F1.

Table 3. Single-split KD sensitivity analysis for the resnet50 teacher (fold 0 of the Cassava cross-validation split). Temperature is swept at $\alpha = 0.7$ and α at $T = 4$; † marks the default used in the main study. Values are single-split point estimates (no fold error bars).

Sweep	T	α	Accuracy (%)	Macro-F1 (%)
temperature 1	0.7		83.50	71.35
temperature 2	0.7		83.79	71.29
temperature 4	0.7 †		84.14	71.52
temperature 8	0.7		83.67	71.37
alpha	4	0.3	82.78	70.61
alpha	4	0.5	83.60	70.89
alpha	4	0.7 †	84.14	71.52
alpha	4	0.9	83.11	69.39

The student is stable across both sweeps: accuracy stays within roughly one point over all settings (82.78--84.14%), with a gentle maximum exactly

at the default $T = 4.0$, $\alpha = 0.7$ used in the main study (84.14%). Very low temperatures, very light teacher weighting ($\alpha = 0.3$), and very heavy teacher weighting ($\alpha = 0.9$) are all slightly worse, and $\alpha = 0.9$ in particular lowers macro-F1 (to 69.39%), consistent with over-weighting the teacher at the expense of the minority classes. The flat response confirms that the fixed configuration used in the main comparison is well justified and that the teacher comparison is not an artifact of a particular hyperparameter choice. These results are single-split point estimates without fold error bars, so they are intended as a sensitivity check rather than as a second cross-validated comparison.

5.3 Edge Compression: Quantization

Distillation reduces the parameter count, but the deployed student can be compressed further with post-training quantization (PTQ). We apply two int8 PTQ strategies to the single-split distilled MobileNetV3-Small student and evaluate them on CPU, the relevant setting for edge inference: *dynamic* PTQ (weights only, no calibration) and *static* PTQ (weights and activations, calibrated on 256 training images). Measurements use the `fbgemm` x86 backend; an ARM backend (`qnnpack`) would be the target for mobile deployment. Following standard PTQ evaluation practice, quantization is applied to a single representative model, whose float32 accuracy (84.2%) lies within one point of the cross-validated mean. Table 4 reports the accuracy/size/latency trade-off.

Table 4. Post-training quantization of the distilled MobileNetV3-Small student on Cassava (single-split test set, CPU, `fbgemm`). Precision, recall, and F1 are macro-averaged.

Variant	Accuracy (%)	Precision (%)	Recall (%)	Macro-F1 (%)	Size (MB)	Latency (ms)
Float32 (baseline)	84.23	74.31	70.42	71.87	5.93	17.7
Dynamic int8	84.20	74.32	70.35	71.83	4.23	16.0
Static int8	12.22	21.06	20.43	6.05	1.74	18.4

Dynamic quantization is essentially lossless (accuracy $84.23 \rightarrow 84.20\%$, a 0.03-point change) but compresses the model only $1.4\times$, because it quantizes the linear layers while leaving the convolution-dominated backbone in float. Static quantization reaches $3.4\times$ compression but collapses accuracy to 12.2%, below the 20% random baseline. This collapse is consistent with the known difficulty of quantizing MobileNetV3: its depthwise-separable convolutions, Hardswish activations, and squeeze-and-excitation blocks are sensitive to post-training quantization, and `torchvision` provides quantization-ready variants only for MobileNetV2 and MobileNetV3-Large, not the Small variant used here. Neither mode improves batch-one CPU latency, because the model is small enough that integer-kernel and quantize/dequantize overheads offset the savings --- echoing the earlier observation that model size and latency are distinct axes. In practice, dynamic PTQ offers a free $1.4\times$ size reduction at no accuracy cost, whereas

reaching the full int8 compression without destroying accuracy would require quantization-aware training (QAT), which we leave to future work.

5.4 Preliminary Study: PlantVillage Single Split

Table 5 reports the preliminary single-split results on the 38-class PlantVillage task. The ResNet50 teacher reaches 0.9936 test accuracy and 0.9912 macro-F1. The supervised MobileNetV3-Small baseline is already strong, with 0.9851 accuracy and 0.9779 macro-F1. After distillation from ResNet50, MobileNetV3-Small improves to 0.9937 accuracy and 0.9907 macro-F1 while retaining the same 1.56M parameter count.

Table 5. Preliminary single-split test results on the 38-class PlantVillage task.

Model	Training strategy	Accuracy	Macro-F1	Parameters	Latency
ResNet50	Teacher supervised	0.9936	0.9912	23,585,894	5.47 ms
MobileNetV3-Small	Supervised baseline	0.9851	0.9779	1,556,806	3.51 ms
MobileNetV3-Small	Distilled from ResNet50	0.9937	0.9907	1,556,806	3.53 ms

The comparison between the supervised and distilled MobileNetV3-Small models is informative because the architecture and parameter count are identical in both rows. The improvement from 0.9851 to 0.9937 accuracy therefore does not come from increasing student capacity; it indicates that the teacher logits provide useful training signal beyond the hard labels alone. At the same time, PlantVillage images are relatively clean and the supervised student is already very strong, which limits the headroom for distillation. This motivates the main cross-validated study on the harder Cassava data, where the supervised baseline is weaker and the effect of the teacher can be observed more clearly.

6 Error Analysis

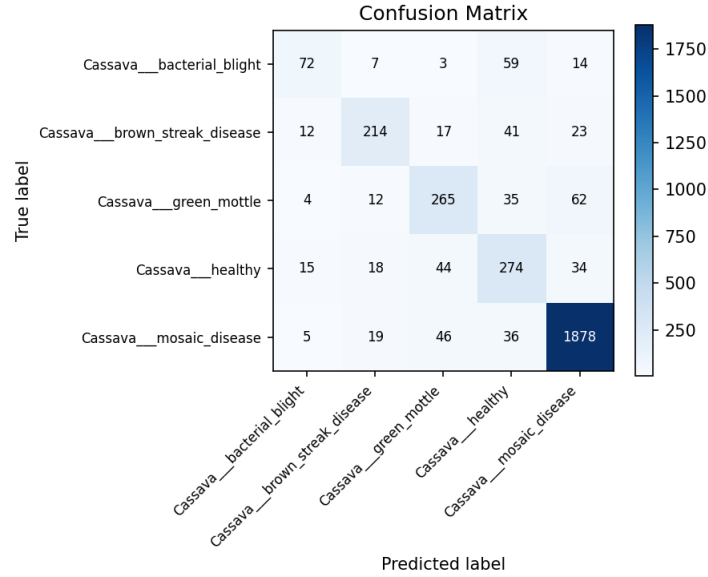
The gap between accuracy and macro-F1 that recurs throughout the Cassava results is a direct consequence of class imbalance. On the single-split test set the five classes are highly skewed: *mosaic disease* accounts for 1,984 of 3,209 images (62%), whereas *bacterial blight* has only 155 (5%). Table 6 gives the per-class performance of the distilled student and Figure 1 its confusion matrix.

The student is excellent on the dominant class (mosaic disease, F1 0.94) but degrades sharply on the minority classes, reaching its worst on bacterial blight (F1 0.55, recall 0.47). Because accuracy is dominated by the majority class while macro-F1 weights all classes equally, accuracy (84%) substantially overstates per-class competence (macro-F1 72%) --- which is why we report both throughout.

The confusion matrix exposes two systematic error patterns. First, and most consequential for a disease-screening application, minority diseases are frequently

Table 6. Per-class performance of the distilled MobileNetV3-Small student on the Cassava single-split test set.

Class	Precision	Recall	F1	Support
Bacterial blight	0.667	0.465	0.548	155
Brown streak disease	0.793	0.697	0.742	307
Green mottle	0.707	0.701	0.704	378
Healthy	0.616	0.712	0.660	385
Mosaic disease	0.934	0.947	0.940	1,984

**Fig. 1.** Confusion matrix of the distilled MobileNetV3-Small student on the Cassava single-split test set. Errors concentrate in the minority classes, which are most often confused with *healthy* and with the dominant *mosaic disease* class.

predicted as *healthy*: 59 of 155 bacterial-blight leaves (38%) and 41 brown-streak leaves are misclassified as healthy. These are false negatives --- diseased plants labelled healthy --- the most costly error type in crop monitoring. Second, the two visually similar mottling classes, *green mottle* and *mosaic disease*, are mutually confused (62 and 46 cases), reflecting genuine visual similarity rather than imbalance alone. Distillation improves overall calibration but inherits this confusion structure rather than resolving the hardest minority cases; addressing them would likely require class re-balancing, targeted augmentation, or cost-sensitive training, which we identify as future work.

7 Discussion

The two studies tell a consistent story. On both PlantVillage and the harder Cassava dataset, distilling a large teacher into MobileNetV3-Small improves the compact student over supervised training at no inference cost. The central finding of the cross-validated study is that this improvement is largely independent of the teacher: three teachers spanning convolutional and transformer architectures, an order of magnitude in size, and several points of accuracy yield distilled students that differ by less than their across-fold variation. This suggests the benefit stems from the softened-label regularization of distillation rather than from any particular teacher, which is practically convenient for edge settings because a small, inexpensive teacher suffices. Quantization provides a further, orthogonal compression: dynamic int8 post-training quantization shrinks the deployed student at no accuracy cost, although the architecture resists more aggressive static quantization without quantization-aware training.

All experiments are configuration-driven. The preliminary PlantVillage runs correspond to the full-dataset teacher, student, and distillation configurations, while the main study is driven by the cross-validation configs and executed with the `src.cross_validate` runner; results are collated into a single table with `src.collate_cv`. Reported metrics are taken from the stored result files rather than copied from terminal logs, so the report stays tied to reproducible artifacts: each run stores a model summary, per-fold metrics, final aggregated metrics, and result CSVs.

Several limitations remain. First, the distillation hyperparameters are kept fixed across teachers ($T = 4.0$, with $\alpha = 0.5$ or 0.7); a more complete study would tune temperature and loss weighting per teacher, which would help separate the effect of the teacher architecture from that of the distillation schedule. Second, the conclusion that the teacher matters little rests on the distilled students differing by no more than their across-fold variation; with only five folds the statistical power is limited, so this is evidence that any teacher effect is small rather than proof that it is exactly zero, and more folds or repeated runs would tighten the comparison. Third, the per-fold teacher retraining means each teacher is trained multiple times, which is computationally expensive but necessary to avoid test leakage. Fourth, latency is measured locally --- on Apple Silicon (MPS) for the preliminary study and on an NVIDIA RTX 4070 (CUDA) for the main study ---

so latency values are comparable only within each study and should not be read as deployment benchmarks. Finally, both datasets are curated; performance on field images with cluttered backgrounds, varying lighting, and different cameras may be lower.

8 Conclusion

This work studied knowledge distillation from large teachers into a compact MobileNetV3-Small student for edge-efficient plant disease classification. A preliminary single-split experiment on the 38-class PlantVillage dataset showed that a ResNet50 teacher can be distilled into a student with 0.9937 accuracy and 0.9907 macro-F1 at about 15 times fewer parameters than the teacher. The main study then used stratified five-fold cross-validation on the harder, imbalanced Cassava dataset to compare three teacher architectures --- ResNet50, EfficientNet-B2, and ViT-B/16 --- distilled into the same student, reporting accuracy together with macro-averaged precision, recall, and F1. On Cassava, distillation raised the student by 1.4--1.8 accuracy points over the supervised baseline, a gain that was statistically significant for the ResNet50 and ViT-B/16 teachers. The choice of teacher, however, had little effect: the three distilled students agreed to within half a point despite teachers that differed widely in accuracy, architecture, and size, and the weakest and largest teacher (ViT-B/16) taught as effectively as the strongest while its student even surpassed it. The benefit of distillation in this setting therefore appears to stem from the procedure rather than from the capacity or architecture of the teacher, which is encouraging for edge deployment because a small, inexpensive teacher suffices. The main limitations are dataset realism and the use of fixed distillation hyperparameters; future work should tune the distillation schedule per teacher, evaluate on more realistic field imagery, and measure latency on actual edge hardware.

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